



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

BT1001-1 – BIOCHEMISTRY

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following polysaccharides is a major structural component of plant cell walls?
A) Glycogen B) Starch
C) Cellulose D) Chitin
2. More complex carbohydrate polymers covalently attached to proteins or lipids, which is biologically active molecules called
A) Glycoconjugates B) Proteoglycans
C) Chitin D) Sulfolipids
3. Which of the following is a major component of proteoglycans?
A) Glycosaminoglycans B) Glycolipids
C) Nucleic acids D) Phospholipids
4. Amylopectin and cellulose are compared and both these polysaccharide _____
A) function as structural components B) contain α - glycosidic bond
C) have extended linear shapes D) contain 1,4-glycosidic bond
5. Sphingolipids contains _____ alcoholic group
A) Ethanol B) Glycerol
C) Sphingosine D) Acetone
6. Which lipid class includes cholesterol and its derivatives?
A) Sterols B) Phospholipids
C) Glycolipids D) Triglycerides
7. Which of the following is a characteristic of saturated fatty acids?
A) They have double bonds between carbon atoms. B) They are typically liquid at room temperature
C) They have no double bonds between carbon atoms D) They contain more oxygen than unsaturated fatty acids
8. The degree of unsaturation of lipid can be measured as
A) Saponification number B) Iodine number
C) Polenske number D) Reichert meissil number
9. What type of bond links amino acids together in a peptide chain?
A) Hydrogen bond B) Ionic bond
C) Peptide bond D) Disulfide bond
10. Which peptide bond property contributes to the rigidity of protein structures?
A) The planar nature of the peptide bond B) The ability to form hydrogen bonds
C) The flexibility of the peptide bond D) The ability to form disulfide bridges
11. Which of the following proteins is classified as a fibrous protein?
A) Hemoglobin B) Enzyme
C) Collagen D) Antibody
12. Which globular protein is responsible for transporting oxygen in the blood?
A) Hemoglobin B) Collagen
C) Elastin D) Fibronectin

P.T.O.

BT1001-1

SEE – December 2025

13. Glycoproteins is linked to oligosaccharides via _____
 A) Covalent bond B) Hydrogen bond
 C) Hydrophobic bond D) Electrostatic bond
14. The enzyme that joins DNA cuts is called _____
 A) Lipase B) Ligase
 C) DNA phosphorylation D) Reverse transcriptase
15. What term describes the tertiary structure of DNA?
 A) Double helix B) Beta-pleated sheet
 C) Supercoiling D) Hairpin loop
16. Which of the following amino acids in histones interact with DNA backbone?
 A) Cys and Met B) Ile and Val
 C) Leu and Ala D) Lys and Arg
17. Where does gluconeogenesis primarily occur in the body?
 A) Muscle B) Brain
 C) Liver D) Adipose tissue
18. What is the primary purpose of the pentose phosphate pathway?
 A) To generate ATP B) To produce NADPH and ribose-5-phosphate
 C) To degrade fatty acids D) To synthesize cholesterol
19. In the beta-oxidation of fatty acids, what is the primary product after the removal of each two-carbon unit?
 A) Propionyl-CoA B) Butyryl-CoA
 C) Acetyl-CoA D) Palmitoyl-CoA
20. End products of pentose phosphate pathways are: _____
 A) Ribose-5 phosphate B) Nucleotides
 C) DNA D) Ribose -2 phosphate

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Define proteoglycans. Explain the structural features of proteoglycans.	8	L*2	1	1
b)	Illustrate structural details of sphingolipids with examples.	8	L2	2	1
2. a)	Depicting the structure, explain how amylopectin and cellulose differs in their structural features. Add a note on its function.	8	L2	1	1
b)	Justify lipids as biological signal molecules.	8	L2	2	1
3. a)	Differentiate homo and heteropolysaccharides with examples. Add a note on glycoproteins.	8	L2	1	1
b)	Write the structure of triglyceride. Add a note on properties of lipids.	8	L2	2	1
	Unit – II				
4. a)	Explain classification of proteins. Add a note on globular proteins.	8	L2	3	1
b)	Define nucleosome and add a note on different levels of complex arrangements of DNA during chromatin organization.	8	L2	4	1
5. a)	Give an account on classification of amino acids.	8	L2	3	1
b)	Elucidate on the secondary structure of DNA.	8	L2	4	1
6. a)	What are peptides? Add a note on tertiary and quaternary structure of proteins.	8	L2	3	1
b)	With diagram, explain secondary structure of t-RNA.	8	L2	4	1
	Unit – III				
7. a)	Explain pentose phosphate pathway. Add a note on its importance.	8	L2	5	1
b)	Explain the β -oxidation of palmitic acid. Outline the steps involved in the conversion of palmitate to acetyl-CoA.	8	L2	5	1
8. a)	Depict the process of glycolysis and mention its significance.	8	L2	5	1
b)	Why TCA cycle is called amphibolic pathway? Illustrate TCA cycle.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
